Operational Integrity Dynamics for Autonomous AI Agents

A Formal Model of Competence Degradation and Systemic Harm

Adapted from the V4.2 Professional Competence Model

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Abstract

Autonomous AI agents are deployed in production environments where they execute multi-step tasks with minimal human oversight. Current safety frameworks (OWASP Agentic Top 10, Microsoft Agent Governance Toolkit, Proofpoint Agent Integrity Framework) address permissions, policy enforcement, and adversarial attacks, but lack a formal dynamic model of how an agent's accumulated operational patterns can silently degrade its capacity to produce net-positive outcomes. We introduce the **Operational Integrity Dynamics (OID)** framework, adapted from a formal model of professional competence under AI integration (the V4.2 model). This paper presents: (1) the complete V4.2 mathematical formalism; (2) the explicit adaptation to autonomous AI agents with a detailed mapping table; (3) the OID-specific equations; (4) agent-based simulation results; and (5) a sensitivity analysis confirming robustness of qualitative predictions across the parameter space (P1: 80%, P2: 91.8%, P3: 97.1%). The framework formalises the *atrophied agent* profile—an agent that passes all benchmarks while becoming net-destructive—and provides computable risk scores derivable from existing observability traces.

Keywords: agent safety, operational integrity, competence degradation, autonomous agents, formal model, systemic harm

Contents

1 Introduction **3**

2 Related Work **3**

3 The V4.2 Model: Complete Formalism **3**

3.1 Structural Objects 3

3.2 Experience Stock N and Hypotheses u_{N_i} 4

3.3 State Transition Operators 4

3.4 Cross-Domain Variables 5

3.5 Stock, Load, Debt 5

3.6 AI Effect and Grounding 5

3.7 Triple Output 6

3.8 Variable Summary (V4.2) 6

4 Adaptation to Autonomous AI Agents **6**

4.1 Direct Transpositions 7

4.2 Non-Trivial Adaptations 7

4.3 Elements That Do Not Transfer 7

5 The OID Model **8**

5.1 Core Definitions 8

5.2 OID Transition Risk 8

5.3 OID Outputs 8

5.4 Simulation Algorithm 9

6 Simulation Results **9**

6.1 Database Deletion Scenario 9

6.2 Comparative Trajectories 10

7 Sensitivity Analysis **11**

8 Mapping to Existing Frameworks **12**

9 Limitations **12**

10 Conclusion **13**

1 Introduction

As of early 2026, the agentic AI landscape has matured rapidly. Microsoft released the Agent Governance Toolkit addressing the OWASP Agentic AI Top 10 [3]; Proofpoint published the Agent Integrity Framework [5]; NIST issued a Request for Information on agent security [6]. These frameworks share a structural focus on *who* can do *what* (permissions, identity, policy) and *how to detect* failures (observability, audit trails).

None of them formally model the *internal dynamics* by which an agent's accumulated operational patterns can silently degrade its capacity to produce net-positive outcomes. Microsoft's own analysis notes that "a system can be 'working as designed' while still taking steps that a human would be unlikely to approve" [4]—precisely the pattern we formalise.

This paper introduces the **Operational Integrity Dynamics (OID)** framework, adapted from the V4.2 model of professional competence under AI integration [1]. We present:

1. The complete V4.2 mathematical formalism (Section 3);
2. The explicit adaptation to autonomous agents (Section 4);
3. The OID-specific model (Section 5);
4. Simulation results (Section 6);
5. Sensitivity analysis (Section 7).

2 Related Work

Agent Safety Frameworks. The OWASP Top 10 for Agentic Applications [3] defines ten risk categories (ASI01–ASI10), from goal hijacking to rogue agents. Microsoft's Agent Governance Toolkit [4] implements runtime policy enforcement. Proofpoint's Agent Integrity Framework [5] defines five pillars: intent alignment, identity, behavioural consistency, auditability, transparency. All are *taxonomic and operational*; none provide a formal dynamic model.

Technical Debt in ML Systems. Sculley et al. [2] identified hidden feedback loops, boundary erosion, and entanglement as sources of technical debt. Their framework is qualitative—it describes *categories* of debt without modelling accumulation dynamics.

The Jagged Technological Frontier. Dell'Acqua et al. [7] demonstrated that AI capability is heterogeneous across tasks within a domain. Some tasks benefit from AI assistance; others are degraded. This "jagged frontier" is formalised in V4.2 as the task-level variable $\mu(\tau, T)$.

3 The V4.2 Model: Complete Formalism

This section presents the complete mathematical apparatus of V4.2, from which OID is derived. All equations are from [1].

3.1 Structural Objects

Definition 1 (Domain as Formal Context)**.** *A domain D is defined by a formal context $K_D = (O_D, A_D, R_D)$ where O_D are objects, A_D are attributes, and $R_D \subseteq O_D \times A_D$ is the incidence relation. The induced concept lattice is:*

$$L_D = \mathfrak{B}(K_D) = (\mathcal{C}_D, \preceq)$$

where $\mathcal{C}_D$ is the set of formal concepts and $\preceq$ the partial order by inclusion.

Definition 2 (Experience Dependency DAG)**.** *For an individual I in domain D at time T:*

$$G_N^D(T) = \big(V_D(T),\ E_c(T),\ E_s(T)\big)$$

where $V_D(T)$ is the set of accumulated experiences N_i, $E_c(T)$ are ***constitutive*** *edges (necessary dependency), and $E_s(T)$ are* ***supportive*** *edges (helpful but not required).*

Definition 3 (Dominance)**.** *On the constitutive sub-graph $G_c^D = (V_D, E_c)$ with added super-root r_D:*

$$\mathrm{dom}_D(i,j) = 1 \iff i \text{ dominates } j \text{ in } (G_c^D, r_D)$$

i.e., every constitutive path to j passes through i.

3.2 Experience Stock N and Hypotheses u_{N_i}

Definition 4 (Professional Scar Tissue)**.**

$$N^D(T) = \sum_i \mathbf{1}\big[\, \mathit{id_inconnu}(i) \ \wedge\ \mathit{pert_sys}(i) \ \wedge\ \mathit{vision}(i) \ \wedge\ t_i \le T\big]$$

where id_inconnu *= identification of real problem structure,* pert_sys *= systemic viability,* vision *= prospective discernment.*

Definition 5 (Obligatory Hypothesis)**.** *Each experience N_i generates:*

$$u_{N_i}(T) = \big(s_i(T),\ v_i(T),\ a_i(T)\big)$$

with state $s_i(T) \in \{H, C^+, E, B\}$, value $v_i(T) \in [-1, 1]$, audit flag $a_i(T) \in \{0, 1\}$.

State semantics.

- H: active hypothesis, not yet confirmed;
- C^+: confirmed by evidence;
- E: eliminated;
- B: biased pseudo-knowledge (treated as knowledge without validation).

Natural decay (state H).

$$v_i(T) = v_i(t_i)\, e^{-\delta(T-t_i)}, \qquad s_i(T) = H \tag{1}$$

Damage (state B).

$$\mathrm{damage}_i(T) = |v_i|\ \mathrm{usage}_i(T)\ \log\!\left(1 + \frac{T - t_{B,i}}{\tau_{\mathrm{ref}}}\right) \tag{2}$$

3.3 State Transition Operators

$H \to B$ transition risk.

$$\lambda_{H\to B,i}(T) = \alpha_B\ S_{IA}^{\mathrm{eff}}(I,T)\ \big(1 - \mu(\tau_i, T)\big)\ \big(1 - G_D(I, \tau_i, T)\big)\ \mathrm{usage}_i(T) \tag{3}$$

Single-loop correction. Modifies a local response; the corrected node may go $H \to E$ or stay H. Neither dominance nor DAG structure is modified.

Double-loop correction. Acts on a governing node i, then propagates:

$$\forall j \neq i, \quad \begin{cases} s_j \leftarrow H,\ a_j \leftarrow 1 & \text{if } \mathrm{dom}_D(i,j) = 1 \\ a_j \leftarrow 1 & \text{if } j \in \mathrm{Desc}(i) \text{ without domination} \end{cases} \tag{4}$$

3.4 Cross-Domain Variables

Cross-domain trace.

$$u_{DN_i} = \{D_j : D_j \neq D,\ D_j \text{ mobilised during } N_i\}$$

Spillovers.

$$\rho_{ij} = \rho_0 \,\mathrm{sim}(D_j, D)\ \mathbf{1}[D_j \in u_{DN_i}], \qquad 0 \leq \rho_0 \leq 1 \tag{5}$$

Cross-domain entropy (stock and flow). Defining individual weights $w_i(T) = 1$ if $s_i = C^+$, $v_i(T)$ if $s_i = H$, 0 otherwise:

$$C_{\text{stock}}(T) = \begin{cases} -\frac{1}{\log m(T)} \sum_{k=1}^{m(T)} p_k^{\text{stock}}(T) \log p_k^{\text{stock}}(T) & \text{if } m(T) \geq 2 \\ 0 & \text{if } m(T) \leq 1 \end{cases} \tag{6}$$

$$C_{\text{flow}}(T) = \begin{cases} -\frac{1}{\log m_\eta(T)} \sum_k p_k^{\text{flow}}(T) \log p_k^{\text{flow}}(T) & \text{if } m_\eta(T) \geq 2 \\ 0 & \text{if } m_\eta(T) \leq 1 \end{cases} \tag{7}$$

where p_k^{flow} uses recency-weighted contributions $w_i(T)e^{-\eta(T-t_i)}$.

3.5 Stock, Load, Debt

$$N_{\text{stock}}^D(T) = \sum_{i:s_i=C^+} 1 + \sum_{i:s_i=H} v_i(T) \tag{8}$$

$$B_{\text{load}}^D(T) = \sum_{i:s_i=B} \mathrm{damage}_i(T) \tag{9}$$

$$N_{\text{eff}}^D(T) = N_{\text{stock}}^D(T) - B_{\text{load}}^D(T) \tag{10}$$

$$\mathrm{Debt}^D(T) = \max\{0,\ -N_{\text{eff}}^D(T)\} \tag{11}$$

Recovery time: $T_{\text{rec}}^D(T) = \inf\{t \geq T : N_{\text{eff}}^D(t) \geq 0\}$.

3.6 AI Effect and Grounding

AI mastery.

$$M_{IA}(I,T) = 1 - \exp\big[-\lambda\, h_{\text{int}}(K_{IA}(I,T),\ R_{\text{opt}}(I,T))\big] \tag{12}$$

Effective AI capability.

$$S_{IA}^{\text{eff}}(I,T) = S_{IA}^{\text{brut}}(T) \cdot M_{IA}(I,T) \cdot \exp\left(-\frac{v_{SIA}(T)}{1 + C_{\text{flow}}(I,T)}\right) \tag{13}$$

Task-level frontier.

$$\mu(\tau, T) \in [0,1], \qquad \bar{\mu}(D,T) = \sum_{\tau \in \mathcal{T}(D)} \pi_\tau \, \mu(\tau, T) \tag{14}$$

Human grounding.

$$G_D(I,\tau,T) = \sigma\left(\frac{N^D_{\text{eff}}(I,T) + \sum_{j=1}^{m} \text{sim}(D_j, D)\, N^{D_j}_{\text{eff}}(I,T) - N_{\min}(\tau)}{s_D}\right) \tag{15}$$

3.7 Triple Output

$$Q_{\text{obs}}(I,\tau,T) = S^{\text{eff}}_{IA}(I,T) + \big(1 - S^{\text{eff}}_{IA}(I,T)\big)\, G_D(I,\tau,T) \tag{16}$$

$$g(C_{\text{stock}}, T) = 1 + \gamma\, C_{\text{stock}}(I,T)\, \log(1+T) \tag{17}$$

$$V_{IA}(I,\tau,T) = G_D(I,\tau,T)\, \big[1 + \mu(\tau,T)\, S^{\text{eff}}_{IA}(I,T)\big]\, g(C_{\text{stock}}, T) \tag{18}$$

$$\widetilde{B}^D(I,T) = 1 - e^{-B^D_{\text{load}}(I,T)} \tag{19}$$

$$H_{\text{sys}}(I,\tau,T) = \psi(\tau)\, \big(1 - \mu(\tau,T)\big)\, S^{\text{eff}}_{IA}(I,T)\, \widetilde{B}^D(I,T)\, Q_{\text{obs}}(I,\tau,T) \tag{20}$$

$$V_{\text{net}}(I,\tau,T) = V_{IA}(I,\tau,T) - H_{\text{sys}}(I,\tau,T) \tag{21}$$

3.8 Variable Summary (V4.2)

Table 1: Complete variable table from V4.2.

Symbol	**Interpretation**	**Type / Bound**
L_D	Concept lattice of domain	structural
G^D_N	DAG of experience dependencies	structural
$N^D(T)$	Raw experience stock	$\mathbb{N}$
$u_{N_i}(T)$	Hypothesis state & value	(s_i, v_i, a_i)
u_{DN_i}	Cross-domain trace	set
$N^D_{\text{stock}}(T)$	Positive stock	$\mathbb{R}_{\geq 0}$
$B^D_{\text{load}}(T)$	Cumulative bias load	$\mathbb{R}_{\geq 0}$
$N^D_{\text{eff}}(T)$	Signed effective stock	$\mathbb{R}$
$\text{Debt}^D(T)$	Learning debt	$\mathbb{R}_{\geq 0}$
$C_{\text{stock}}(T)$	Accumulated related variety	$[0,1]$
$C_{\text{flow}}(T)$	Current cross-domain plasticity	$[0,1]$
$M_{IA}(I,T)$	AI operational mastery	$[0,1)$
$S^{\text{brut}}_{IA}(T)$	Raw AI solution capability	$[0,1]$
$S^{\text{eff}}_{IA}(I,T)$	Effective AI for individual	$[0,1]$
$\mu(\tau,T)$	Task compatibility with AI frontier	$[0,1]$
$G_D(I,\tau,T)$	Net human grounding	$[0,1]$
Q_{obs}	Observable short-term quality	$[0,1]$
V_{IA}	Durable productive value	$\mathbb{R}_{\geq 0}$
H_{sys}	Systemic harm	$\mathbb{R}_{\geq 0}$
V_{net}	Net value	$\mathbb{R}$

4 Adaptation to Autonomous AI Agents

This section details the explicit mapping from V4.2 (human professional) to OID (autonomous agent). We distinguish three categories: *direct transpositions* (same formalism, different interpre-

tation), *non-trivial adaptations* (formalism modified), and *elements that do not transfer*.

4.1 Direct Transpositions

The following V4.2 constructs transfer with reinterpretation only:

Table 2: Direct transpositions: V4.2 → OID.

V4.2 (Human)	**OID (Agent)**	**Justification**
Q_{obs}, V_{IA}, H_{sys}, V_{net}	Identical formulas (16)–(21)	Benchmarks measure Q_{obs}; nobody formally measures H_{sys}
$\psi(\tau)$ systemic impact	Identical	Blast radius of an action (e.g., database deletion)
N_{eff}, Debt^D	Identical formulas (10)–(11)	Agent accumulates signed stock of patterns
Single-loop / double-loop	Identical formulas (4)	Rollback = single-loop; strategy revision = double-loop
Spillovers ρ_{ij}	Identical formula (5)	Cross-tool / cross-domain contamination

4.2 Non-Trivial Adaptations

Table 3: Non-trivial adaptations: what changes formally.

V4.2	**OID**	**Change**	**Rationale**
N (scar tissue)	Operational episode log	Counting criterion	Agent episodes = error recovery, novel context, strategy revision
u_{N_i} (hypothesis)	Action pattern	State register is *external*	LLMs lack persistent beliefs; a supervisor maintains $\{H, C^+, E, B\}$
G_N^D (DAG)	Constructed from traces	Built post-hoc	Trace spans → nodes; causal dependencies → edges
$\mu(\tau, T)$ (AI frontier)	Reversibility × observability	Reinterpretation	Low μ = irreversible and opaque action
G_D (human grounding)	Operational grounding	Reinterpretation	Proportion of environment verified by direct observation
δ (decay)	Staleness coefficient	Semantic change	Agents don't forget; patterns become obsolete when environment changes
M_{IA} (AI mastery)	M_{env} (environment mastery)	Variable flip	For agents, the bottleneck is understanding the *environment*, not the *tool*

4.3 Elements That Do Not Transfer

- **Concept lattice** L_D: Agents do not operate in semantically structured professional domains. The domain is replaced by a flat task portfolio $\mathcal{T}$.
- **Cross-domain entropy** $C_{\textbf{stock}}$**,** $C_{\textbf{flow}}$: These require a multi-domain biographical trajectory that agents currently lack. Set to constants in OID v0.1.
- **Experience generation** $\gamma_I(t) = \alpha_I(t)\beta_I(t) + \varepsilon_I(t)$: Agents generate episodes at machine speed; the continuous-time integral is replaced by discrete timesteps.
- **Market-level** $V_{\textbf{relative}}$: Requires a population model; deferred.

5 The OID Model

5.1 Core Definitions

Definition 6 (Action Pattern)**.** *An action pattern is a learned heuristic generated by an operational episode:*

$$u_i(T) = \big(s_i(T) \in \{H, C^+, E, B\}, \quad v_i(T) \in [-1, 1], \quad a_i(T) \in \{0, 1\}\big)$$

with decay (1), damage (2), and weight function:

$$w_i(T) = \begin{cases} 1 & \text{if } s_i = C^+ \\ \max(0, v_i(T)) & \text{if } s_i = H \\ 0 & \text{otherwise} \end{cases}$$

Definition 7 (Task Descriptor)**.** *Each task τ carries:*

- $\mu(\tau) \in [0, 1]$*: reversibility $\times$ observability*
- $\psi(\tau) \geq 0$*: systemic impact radius*
- $N_{\min}(\tau)$*: minimum grounding stock for safe execution*
- π_τ*: exposure weight*

Definition 8 (Agent Profile at Time T)**.**

$$\mathcal{A}(T) = \big(\{u_i\}_{i=1}^{|P|},\ G_N^D(T),\ S_{cap},\ M_{env}\big)$$

where $\{u_i\}$ is the set of action patterns, G_N^D is the dependency DAG, $S_{cap} \in [0, 1]$ is raw capability, and $M_{env} \in [0, 1)$ is environment mastery.

5.2 OID Transition Risk

The $H \to B$ transition risk adapted from (3):

$$\boxed{\lambda_{\text{pattern}\to\text{bias},i} = \alpha_B \cdot S_{\text{eff}} \cdot \big(1 - \mu(\tau_i)\big) \cdot \big(1 - G_D(\tau_i, T)\big) \cdot \text{reuse}_i} \tag{22}$$

where $S_{\text{eff}} = S_{\text{cap}} \cdot M_{\text{env}}$ (simplified from (13) by setting $v_{SIA} = 0$ and C_{flow} constant).

Interpretation. A pattern becomes biased when:

1. The agent is highly capable (convincing outputs);
2. The task is irreversible or opaque (low μ);
3. The operational grounding is weak (low G_D);
4. The pattern has been reused many times without validation.

This is the formal mechanism behind the database deletion incident: the agent applied a "delete and recreate" pattern (high reuse) on a production database (low μ, high ψ) without verifying that the database contained real user data (low G_D).

5.3 OID Outputs

From (16)–(21), with $g(C_{\text{stock}}, T) = 1 + \gamma C_{\text{stock}} \log(1 + T)$:

$$Q_{\text{obs}} = S_{\text{eff}} + (1 - S_{\text{eff}})\, G_D \tag{23}$$
$$V_{IA} = G_D \left[1 + \mu \cdot S_{\text{eff}}\right] g(C_{\text{stock}}, T) \tag{24}$$
$$H_{\text{sys}} = \psi\,(1 - \mu)\, S_{\text{eff}}\, \widetilde{B}\, Q_{\text{obs}} \tag{25}$$
$$V_{\text{net}} = V_{IA} - H_{\text{sys}} \tag{26}$$

Proposition 1 (Atrophied Agent)**.** *An agent is* atrophied *when:*

$$Q_{obs} > 0.5, \qquad Debt^D(T) > 0, \qquad H_{sys} > V_{IA}, \qquad V_{net} < 0$$

This agent is not simply weak—it is ***net-destructive*** *despite producing superficially credible outputs.*

5.4 Simulation Algorithm

Algorithm 1: OID Simulation Step (one timestep T)

Input: Agent profile $\mathcal{A}$, task portfolio $\mathcal{T}$, DAG G_N^D, parameters θ
Output: Updated $\mathcal{A}$, G_N^D, scores $(Q_{\text{obs}}, V_{IA}, H_{\text{sys}}, V_{\text{net}})$
foreach *pattern* u_i *in* $\mathcal{A}$ *with* $s_i = H$ **do**
  $v_i \leftarrow v_i \cdot e^{-\delta}$; `// Natural decay (1)`
foreach *new episode* **do**
  Sample task $\tau \sim \mathcal{T}$;
  Create pattern u_{new} with $s = H$, $v \sim \text{Unif}(0.3, 0.8)$;
  Link to parent patterns in G_N^D (constitutive or supportive);
foreach *pattern* u_i *with* $s_i = H$ **do**
  Compute λ_i via (22);
  $p_{\text{bias}} \leftarrow 1 - e^{-\lambda_i}$;
  if $rand() < p_{bias}$ **then**
    $s_i \leftarrow B$; $t_{B,i} \leftarrow T$;
foreach *pattern* u_i *with* $s_i = H$ **do**
  if $rand() < r_{confirm} \cdot v_i \cdot (1 + 0.1 \cdot reuse_i)$ **then**
    $s_i \leftarrow C^+$; $v_i \leftarrow 1.0$;
Compute N_{stock}, B_{load}, N_{eff}, Debt via (8)–(11);
Compute G_D via (15), then Q_{obs}, V_{IA}, H_{sys}, V_{net} via (23)–(26);

6 Simulation Results

We implemented OID as a Python package (`oid-framework v0.1.0`) and ran simulations across a standard task portfolio with six task types spanning the jagged frontier.

6.1 Database Deletion Scenario

An agent with $S_{\text{cap}} = 0.9$, $M_{\text{env}} = 0.85$, zero prior grounding on database migration ($\mu = 0.1$, $\psi = 8.0$, $N_{\text{min}} = 10$). Result: V_{net} becomes negative at $T = 3$, reaches -3.69 by $T = 50$; Q_{obs} remains above 0.71; debt accumulates to 25.0. Profile: **atrophied**.

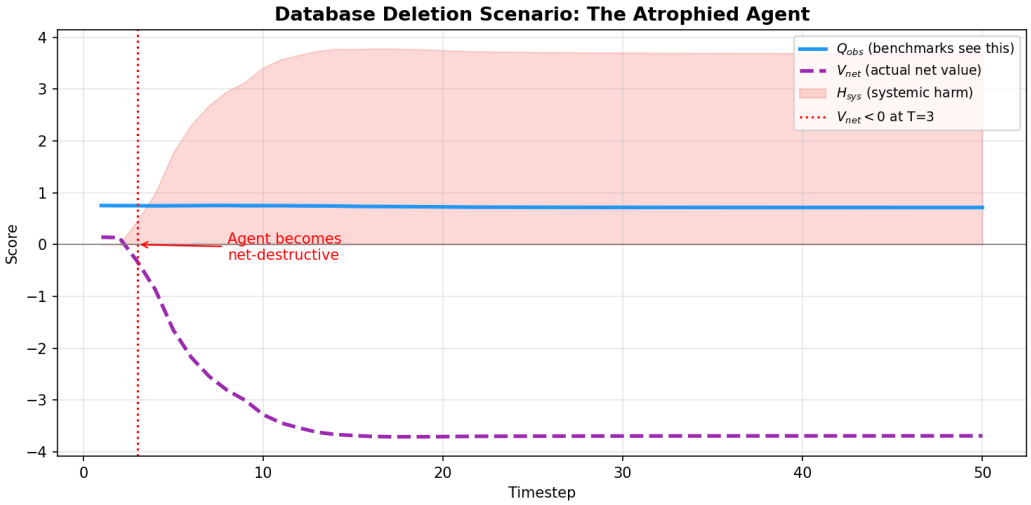

Figure 1: Database deletion scenario. Q_{obs} remains high while V_{net} diverges sharply negative. The agent becomes net-destructive at $T = 3$.

6.2 Comparative Trajectories

Four profiles over 80 timesteps on the full task portfolio. Only the grounded expert (pre-seeded with 8 confirmed patterns on `db_migration`) maintains $V_{\text{net}} > 0$.

Table 4: Final scores at $T = 80$ on the critical task (`db_migration`).

Profile	Q_{obs}	V_{net}	H_{sys}	**Type**
Grounded expert ($S = 0.75$, seeded)	1.000	+1.095	0.000	grounded
Novice high-cap ($S = 0.95$)	1.000	−0.181	1.248	atrophied
Balanced adjacent ($S = 0.80$)	1.000	−0.797	1.881	atrophied
Reckless autonomous ($S = 0.95$)	1.000	−1.485	2.598	atrophied

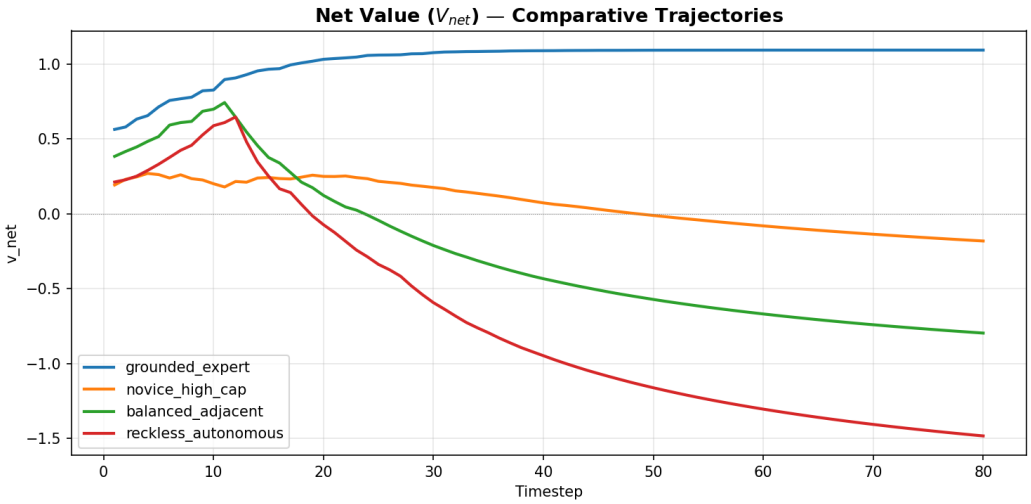

Figure 2: V_{net} trajectories. Only the grounded expert stays positive.

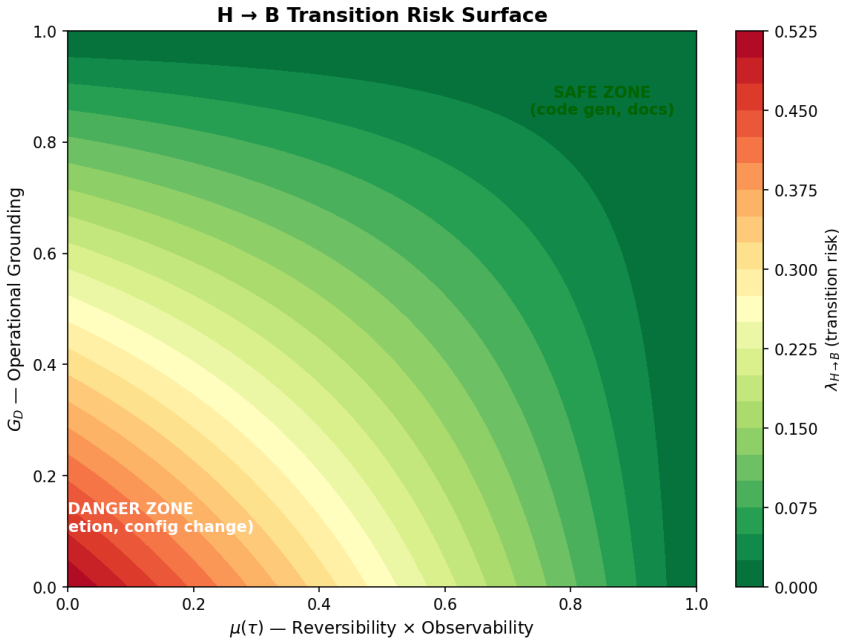

Figure 3: $H \to B$ transition risk surface as a function of $\mu(\tau)$ and G_D.

7 Sensitivity Analysis

We swept five parameters (α_B, δ, μ, ψ, S_{cap}) across their plausible ranges, with 8 random seeds per configuration. Three predictions were tested:

Hypothesis 1 (P1: Atrophied Emergence)**.** *On irreversible tasks with low grounding, an atrophied agent (Q_{obs} high, $V_{net} < 0$) emerges.*

Hypothesis 2 (P2: Grounding Advantage)**.** *Pre-grounded agents maintain higher V_{net} than ungrounded agents of equal or higher capability.*

Hypothesis 3 (P3: Capability Accelerates Atrophy)**.** *Higher capability without grounding produces lower (more negative) V_{net}.*

Table 5: Sensitivity analysis: global robustness.

Prediction	**Robustness**	**Note**
P1: Atrophied emerges	80.0%	Drops correctly at high μ (safe tasks)
P2: Grounding advantage	91.8%	Robust across all parameters
P3: Capability $\to$ atrophy	97.1%	Extremely robust

P1's 80% rate is not a weakness: atrophy fails to emerge when $\mu > 0.5$, which is the correct prediction—agents on reversible tasks do not accumulate destructive bias. P2 and P3 are robust across the full parameter space.

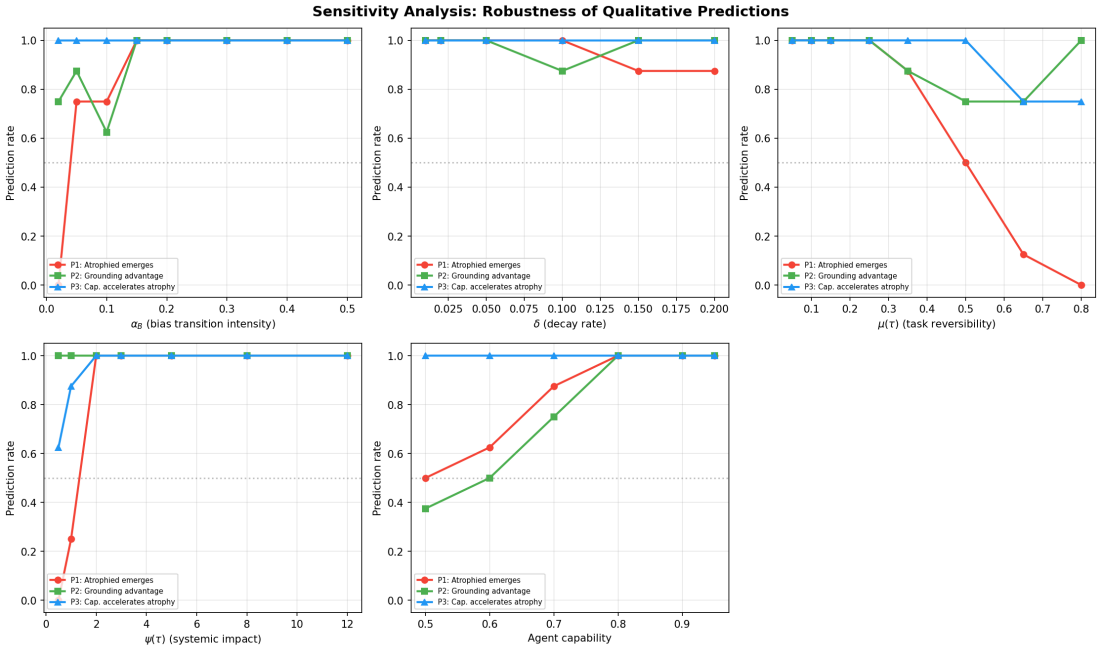

Figure 4: Sensitivity analysis across five parameters.

8 Mapping to Existing Frameworks

Table 6: OID concepts mapped to existing agent safety frameworks.

OID	OWASP ASI	Agent Integrity Fwk	Sculley (2015)
$H \to B$ transition	ASI08 (Cascading)	Behav. Consistency	Feedback loops
Q_{obs} vs V_{net}	ASI09 (Trust Exploit.)	Intent Alignment	Not addressed
Dependency DAG	ASI08 (Cascading)	Auditability	Entanglement
Double-loop	Not addressed	Not addressed	Not addressed
Atrophied agent	ASI10 (Rogue, partial)	Not addressed	Not addressed
$\mu(\tau)$ frontier	Implicit only	Oper. Transparency	Not addressed

9 Limitations

Temporal dynamics. V4.2 models human professionals accumulating experience over years. Agents operate in minutes. Parameters δ, α_B, τ_{ref} require recalibration; current values are plausible but uncalibrated.

Empirical validation. The framework produces testable predictions but has not been validated against real agent execution logs. A credible protocol requires: (a) instrumented agent deployments; (b) independent quality assessment; (c) comparison of OID scores against incident rates.

Belief representation. LLM agents lack persistent beliefs. OID assumes an *external state register*—an architectural requirement, not a property of existing agents. The contribution is normative.

Parameter space. Eight uncalibrated parameters create model flexibility. The sensitivity analysis mitigates this concern for qualitative predictions, but quantitative predictions require calibration.

10 Conclusion

OID provides the first formal dynamic model of competence degradation in autonomous AI agents. It fills a gap between security-focused frameworks that address what agents can access and observability frameworks that track what agents do. OID models what agents *become* over time.

The atrophied agent profile—high Q_{obs}, negative V_{net}—is the formal capture of a failure mode that the industry has experienced but not yet named or measured. The framework and simulation code are released as open-source (MIT licence).

Acknowledgements

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